STREAM: sequence-conditioned flow matching for developmental single-cell dynamics

Abstract

STREAM learns continuous developmental vector fields from time-resolved single-cell atlases while conditioning each gene's predicted velocity on linked regulatory DNA sequence. The model combines multi-timepoint conditional flow matching, minibatch optimal transport between neighboring stages, gene-linked candidate cis-regulatory elements, AlphaGenome sequence embeddings, and optional Universal Cell Embedding state representations. The predicted velocity and loss are always expression-valued. In the causal formulation, the frozen UCE encoder is recomputed from the current expression state, defining the autonomous field $dx/dt = f_\theta(E(x), \mathrm{CRE})$. Mouse conditional velocity diagnostics improve when the modeled panel expands from 1,984 to 10,000 genes. We distinguish these path-conditioned diagnostics from genuine held-out-stage forecasting, which transports observed source cells without using the destination state and scores the resulting endpoint distribution.

1 Introduction

Time-resolved single-cell atlases observe development as a sequence of population snapshots. A useful dynamical model should turn those snapshots into a continuous vector field that predicts how cell states move between stages, while remaining comparable across genes, timepoints, and species. Conditional flow matching provides a simulation-free way to fit such vector fields by regressing target velocities along simple paths between empirical distributions [1, 2]. For developmental data, the natural paths are between neighboring timepoints.

STREAM extends this setup by conditioning each gene's velocity on local regulatory sequence. For every selected gene, nearby candidate cis-regulatory elements are embedded with AlphaGenome [6], packed as a short regulatory token sequence, and processed by a shared gene-level transformer. The global cell state conditions these regulatory tokens through either FiLM or cross-attention. The readout from the promoter token is a scalar velocity for that gene. Evaluating the shared module across all selected genes gives an expression-space vector field.

This paper reports the current implementation and benchmarks. The main questions are whether modeling more genes helps held-out timepoint prediction, whether UCE improves the conditioning state, and whether a mouse-trained STREAM vector field can generalize to zebrafish after replacing gene tokens and regulatory sequence inputs.

2 Results

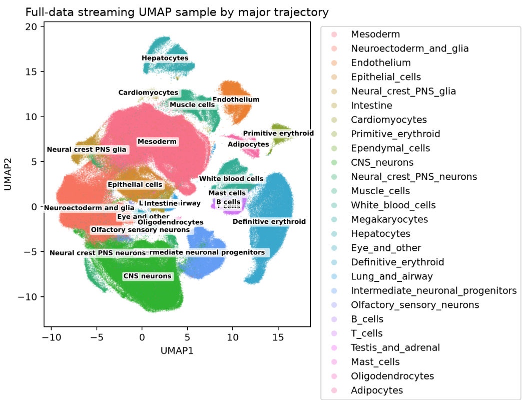
2.1 STREAM predicts expression-space gene velocities from cell state and regulatory sequence

STREAM uses two inputs at each CFM interpolation point: a cell-state vector and a fixed set of regulatory tokens for each gene. The cell-state vector is either expression on the selected gene panel or a frozen 1,280-dimensional UCE embedding computed from that expression state.

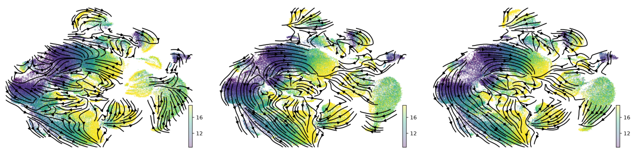
expression endpoints $x_k,\ x_{k+1}$	**minibatch OT pair** (x_k, x_{k+1})	**interpolated state** x_τ or $E(x_\tau)$	**expression velocity** $(x_{k+1} - x_k)/\Delta t$
linked cCREs $\mathcal{C}_g$ within 100 kb	**AlphaGenome embeddings** sequence token vectors	**shared CRE transformer** FiLM or cross-attention by state	**promoter readout** $\hat{v}_g$

Key point: UCE changes the state representation supplied to the network; STREAM still returns one expression velocity per gene after the fixed inverse gene-scale transform.

Figure 1: **STREAM model structure.** Adjacent timepoints are paired by minibatch optimal transport. The model observes an interpolated cell state and gene-specific regulatory tokens, then predicts an expression-space velocity for each gene.



A. Mouse atlas by major trajectory.



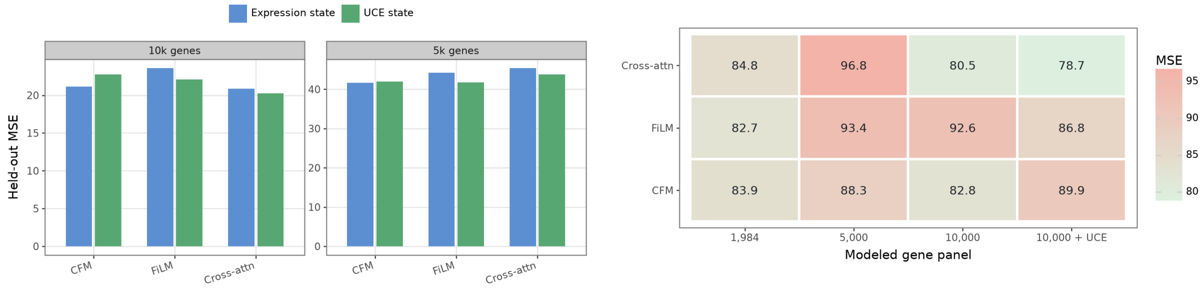
B. Velocity streams for CFM, FiLM STREAM, and cross-attention STREAM.

Figure 2: **Mouse developmental data and learned velocity fields.** STREAM is trained by conditional flow matching, while causal rollout and velocity streams provide qualitative checks on the direction field over the atlas manifold.

The reported output is always a G-dimensional expression velocity, where G is the number of modeled genes (Figure 1). The current coordinate contract fits and integrates gene-scaled expression $Y = Q^{-1}X$, then returns $\mathrm{d}X/\mathrm{d}t = Q\,\mathrm{d}Y/\mathrm{d}t$ for reporting and endpoint evaluation.

2.2 Mouse held-out stages test temporal interpolation across a large developmental atlas

The mouse benchmark uses the public single-cell transcriptional time-lapse of mouse prenatal development [3], represented locally as 11,441,407 cells over 24,552 genes and 43 stages from E8.5 to P0. We held out E9.5 and E10.5 as whole timepoints. Training used only adjacent intervals whose two endpoints were observed training stages. Causal evaluation transports cells from E9.25 to E9.5 and from E10.25 to E10.5 without observing the destination state until scoring. The atlas spans major developmental trajectories, and learned vector fields can be visualized as velocity streams on the shared UMAP (Figure 2).



A. Full modeled panel.

B. Original 1,984-gene panel.

Figure 3: **Mouse conditional velocity fit improves at 10k genes.** Bars and heatmap entries report mean velocity MSE on OT paths from intervals touching held-out stages. These diagnostics condition on both endpoints. Lower is better.

Table 1: **Best 10k mouse conditional-fit results.** Metrics are averaged across OT paths in intervals touching E9.5 or E10.5 and are not rollout scores.

Model	MSE	MAE
10k UCE cross-attention STREAM	20.28	1.594
10k expression cross-attention STREAM	20.86	1.742
10k expression CFM	21.16	1.602
10k UCE FiLM STREAM	22.09	1.629
10k expression FiLM STREAM	23.60	1.669

2.3 Modeling 10k genes improves conditional velocity fit

The original STREAM runs used a 1,984-gene panel. We reran the benchmark with 5,000 and 10,000 protein-coding highly variable genes and evaluated each model on both its full modeled panel and the original 1,984-gene panel. Full-panel conditional velocity losses decreased when moving from 5k to 10k genes for all model families (Figure 3A). On the 10k full panel, cross-attention STREAM slightly improved over expression-only CFM by MSE (20.86 versus 21.16), and adding UCE lowered cross-attention STREAM further to 20.28 MSE and 1.594 MAE. These values evaluate interpolation points constructed from both endpoints and are model-fit diagnostics rather than missing-stage forecasts. Table 1 summarizes the strongest 10k mouse configurations.

The fair legacy-panel comparison asks whether a larger model helps on the same 1,984 genes used by the original run. The 10k cross-attention model improved legacy-panel MSE from 84.79 to 80.54, and 10k UCE cross-attention improved it to 78.75 (Figure 3B). The 5k models did not improve the legacy-panel metric, suggesting that the benefit is not monotonic with panel size and appears at the 10k scale in the current setup.

2.4 Zebrafish transfer shows regime-dependent learned generalization

We next tested cross-species transfer on ZSCAPE, the zebrafish single-cell atlas of perturbed embryos [4]. The active target data retain all 28C control cells with labels beginning `ctrl-`, giving 1,232,014 control cells before excluding cells without a modeled stage label. Modeled stages span 18–96 hpf. We held out 36 and 72 hpf and scored intervals touching those stages.

The transfer benchmark used the best mouse architecture: 10k-gene, UCE cross-attention STREAM. We compared a frozen mouse checkpoint applied zero-shot to zebrafish, the same checkpoint fine-tuned on zebrafish, and a zebrafish-only model. Cross-species scoring used

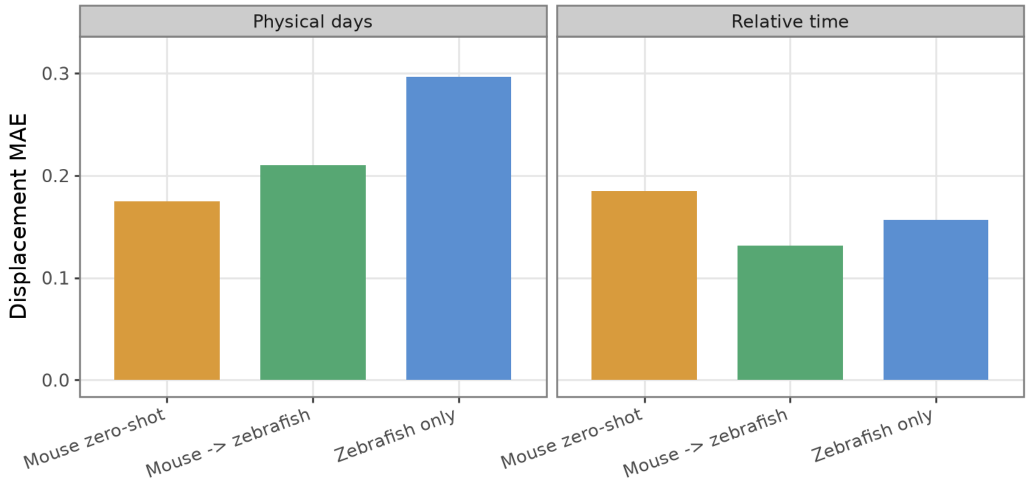


Figure 4: **Zebrafish learned transfer results.** The plot compares only learned models: mouse zero-shot, mouse-initialized fine-tuning, and zebrafish-only training. Lower displacement MAE is better.

Table 2: **Zebrafish cross-species learned-model evaluation.** Held-out stages are 36 and 72 hpf.

Time coordinate	Training regime	Displacement MAE	Displacement MSE
Physical days	Mouse zero-shot	0.175	3.337
Physical days	Mouse $\rightarrow$ zebrafish	0.210	3.351
Physical days	Zebrafish only	0.297	3.414
Relative time	Mouse zero-shot	0.184	3.348
Relative time	Mouse $\rightarrow$ zebrafish	0.131	3.325
Relative time	Zebrafish only	0.157	3.331

expression displacement MAE/MSE, $|(f_\theta - \hat{f})\Delta t|$, so physical-day and relative-time coordinates can be compared on the same expression scale.

Relative-time fine-tuning was the strongest learned zebrafish model (displacement MAE 0.131), outperforming zero-shot (0.184) and zebrafish-only training (0.157). Under physical-day timing, the frozen mouse checkpoint had the lowest learned-model displacement MAE (0.175), followed by fine-tuning (0.210) and zebrafish-only training (0.297) (Figure 4; Table 2). These results indicate a transfer signal, but also show that cross-species alignment is sensitive to the time coordinate and training regime.

3 Methods

3.1 Data sources and regulatory annotations

Table 3 summarizes the expression atlases, gene models, genome references, and cCRE annotations used by the active analyses.

Mouse cCREs were taken from the ENCODE/SCREEN Registry of candidate cis-regulatory elements [7, 8]. Zebrafish cCREs were taken from the ZEPA developmental chromatin accessibility supplement ("cCREs in ZEPA and dynamic cCREs during zebrafish development") and converted from workbook coordinates to BED format [9, 10]. The dynamic zebrafish workbook did not expose readable coordinate sheets in the current conversion, so the final active BED is the primary ZEPA cCRE catalog.

Table 3: **Data sources used by STREAM.** Counts describe the active analysis artifacts after project preprocessing.

Species	Single-cell data	Gene annotation and genome	cCRE source
Mouse	JAX/Shendure mouse prenatal single-cell transcriptional time-lapse; 11,441,407 cells, 24,552 genes, 43 stages from E8.5 to P0	Ensembl GRCm38.95 gene models; mm10/GRCm38 genome sequence	ENCODE/ SCREEN mouse cCRE registry in mm10; 926,843 cCRE rows
Zebrafish	ZSCAPE 28C control embryos; 1,232,014 control cells retained before stage filtering; modeled stages from 18 to 96 hpf	Ensembl Danio rerio GRCz11 release 113 gene models and genome sequence	ZEPA developmental cCRE supplement converted to GRCz11 BED; 959,040 primary cCRE rows

For each species, protein-coding genes were selected by streaming highly variable gene ranking and then matched to the species GTF. Mouse experiments used the original 1,984-gene panel and rerun panels of 5,000 and 10,000 genes. Zebrafish transfer used 10,000 genes. cCREs were linked to a gene when their midpoint was within 100 kb of the gene transcription start site. Positive-strand TSS coordinates used the gene start; negative-strand coordinates used the gene end. The closest cCRE within 1 kb of the TSS was marked as the promoter token. If no such cCRE existed, a 512 bp synthetic promoter window centered on the TSS was inserted. Up to 32 CRE/promoter tokens were packed per gene. The 10k mouse panel produced 317,752 linked CRE/promoter tokens; the 10k zebrafish panel produced 318,223.

3.2 AlphaGenome sequence embeddings

Each real or synthetic CRE window was embedded once with the local PyTorch AlphaGenome implementation. The model used 131,072 bp sequence windows and pooled the 128 bp trunk features to a 3,072-dimensional token vector. Training consumes cached token tensors and does not call AlphaGenome online. The published AlphaGenome model is calibrated for human and mouse organism conditioning. Zebrafish GRCz11 sequence windows therefore used the mouse organism index, which should be treated as an out-of-distribution assumption for cross-species transfer.

3.3 Multi-timepoint conditional flow matching

Let $x \in \mathbb{R}^G$ be expression over the selected gene panel and let $p_k(x)$ be the empirical distribution at observed timepoint t_k. Training is defined over adjacent intervals (t_k, t_{k+1}). Let $q > 0$ contain training-only per-gene count scales, $Q = \mathrm{diag}(q)$, and $y = Q^{-1}x$. For a sampled pair (x_k, x_{k+1}) and $\tau \sim \mathcal{U}(0, 1)$,

$$y_\tau = (1-\tau)y_k + \tau y_{k+1}, \tag{1}$$

$$\hat{v}_Y = \frac{y_{k+1} - y_k}{t_{k+1} - t_k}. \tag{2}$$

The model is trained by minimizing

$$\mathcal{L}(\theta) = \sum_k \mathbb{E}_{\tau,(x_k,x_{k+1})} \left[\left\| f_\theta(s_\tau, \{\mathbf{Z}_g\}_{g=1}^G) - \hat{v}_Y \right\|_2^2 \right], \tag{3}$$

where $s_\tau = y_\tau$ for expression-state runs and $s_\tau = E(Qy_\tau)$ for online-UCE runs. Training pairs are sampled from an entropic minibatch optimal transport plan between cells at the two interval endpoints. The OT cost remains in count-derived expression or train-only PCA space. The fixed transform is inverted before UCE, PCA/Sinkhorn scoring, and biological reporting. This preserves nonnegativity and avoids stochastic reconstruction of counts.

3.4 UCE-state conditioning

For UCE runs, a frozen 33-layer UCE encoder maps the current expression state to $E(x) \in \mathbb{R}^{1280}$ [5]. After expression-space OT pairing and interpolation, the model input is

$$u_\tau = E\left(Q[(1-\tau)y_k + \tau y_{k+1}]\right). \tag{4}$$

The target, prediction, and loss are gene-scaled, while the reported velocity is $Q\hat{v}_Y$. During rollout, UCE is recomputed after every update using $E(Qy)$, yielding the closed autonomous system $\mathrm{d}y/\mathrm{d}t = f_\theta(E(Qy), \{\mathbf{Z}_g\})$. No destination embedding or explicit time input is supplied.

3.5 STREAM architecture and baselines

For gene g, the model receives token embeddings $\mathbf{Z}_g = \{z_{gm}\}_{m=1}^{M}$, signed TSS-relative distances, promoter indicators, and padding masks. The promoter token is always placed first. Token features are projected to model width 256, augmented with signed-distance and promoter embeddings, and processed by a shared 4-layer transformer with 8 attention heads, RoPE positional encoding, and dropout 0.1. We evaluated two state-conditioning mechanisms:

- FiLM: a state MLP produces layer-specific feature-wise scale and shift vectors applied to regulatory token states.
- Cross-attention: a state MLP produces 8 context tokens per layer, and regulatory tokens attend to these context tokens after CRE self-attention.

The scalar velocity for gene g is read from the final promoter-token state. Applying the same gene-level module across all genes forms $f_\theta(s_\tau, \{\mathbf{Z}_g\}_{g=1}^{G}) \in \mathbb{R}^G$. The expression-only CFM baseline uses the same selected genes, same OT sampler, same interpolation path, and same loss, but replaces sequence-conditioned gene modules with an MLP from cell state to expression velocity.

3.6 Relative population growth

To separate state transport from changing lineage abundance, an optional growth head maps the current frozen UCE embedding to one scalar rate per cell, $\gamma_\psi(E(x))$. During unpaired population rollout, cell states follow the fitted ODE or SDE while normalized particle weights satisfy

$$w_{k+1,i} \propto w_{k,i} \exp\left([\gamma_\psi(E(x_{k,i})) - \sum_j w_{k,j}\gamma_\psi(E(x_{k,j}))]\Delta t \right). \tag{5}$$

Centering and normalization identify only relative growth, appropriate because stage-specific atlas sampling does not measure absolute embryo cell number. The endpoint Sinkhorn, gene-mean, and PCA-covariance terms use the predicted weights. Rate and weight-concentration penalties prevent particle collapse. Evaluation reports fixed-mass velocity, growth-only reweighting, and joint velocity-plus-growth controls.

3.7 Held-out timepoint evaluation

Mouse models held out E9.5 and E10.5. Zebrafish models held out 36 and 72 hpf. All intervals touching a held-out endpoint were excluded from training. Conditional-fit diagnostics reuse seeded OT paths and report velocity or expression-displacement error, but are not forecasts because those paths use both endpoints. The causal mouse benchmark instead uses only observed-to-held-out intervals, repeatedly recomputes $E(x)$ during projected Euler integration, and reserves held-out cells for endpoint scoring. It reports train-fitted PCA Sinkhorn divergence, skill relative to source-state persistence, gene-level mean-shift R^2, and mean per-gene Wasserstein distance.

4 Discussion

The current conditional-fit results support the conclusion that larger gene panels are useful at 10k genes: the improvement is visible on each model's full output panel and persists on the original 1,984-gene panel for cross-attention STREAM. The earlier cached-UCE ablations and mouse-to-zebrafish results interpolated endpoint embeddings and therefore measure conditional fit rather than causal forecasting. They should not be used to claim held-out-stage or cross-species predictive generalization until repeated with online UCE.

The main limitations are the simple TSS-window CRE linking rule, the lack of zebrafish-specific AlphaGenome calibration, and the short-interval held-out endpoint metric. Future work should test cell-type-aware regulatory links, sequence embeddings trained or calibrated for zebrafish, and causal rollout evaluation across both species.

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